

1 Introduction and Background

X-ray scattering produces a 1-D intensity profile $I(q)$ that encodes the 3-D shape of a molecule, but computing it exactly via the Debye equation scales as $O(N^2)$ in atom count, making it prohibitively slow for large structures and incompatible with gradient-based optimization. This proposal seeks to answer the question *can a graph neural network learn to predict $I(q)$ directly from atomic coordinates?* To answer this, this project aims to train a message-passing GNN as a fast, differentiable surrogate for the exact computation.

Graph neural networks for molecular property prediction were pioneered by SchNet [1], which introduced continuous-filter convolutions on interatomic distances and remains a strong baseline for scalar molecular properties. DimeNet [2] extended this by incorporating bond angles, improving accuracy on quantum-chemical targets such as the QM9 benchmark [3]. On the scattering side, CRY SOL [4] established the spherical-harmonic decomposition approach that we use to generate ground-truth $I(q)$ labels. More recently, Schütt et al. [5] introduced PaiNN, an equivariant message-passing network.

When a beam of X-rays is fired at a molecule, each atom deflects (“scatters”) the beam by an amount that depends on the atom’s identity and the deflection angle q . Each atom has a characteristic *atomic form factor* $f_k(q)$ encoding how strongly it scatters at angle q . By measuring the total scattered intensity across all angles, we obtain a *scattering profile* $I(q)$ which is a unique fingerprint of the molecule’s 3-D shape [6]. The exact formula, the **Debye equation**, sums the pairwise interatomic interactions:

$$I(q) = \sum_{\alpha} \sum_{\beta} f_{\alpha}(q) f_{\beta}^{*}(q) \operatorname{sinc}(q |\mathbf{r}_{\alpha} - \mathbf{r}_{\beta}|) \quad (1)$$

where $\operatorname{sinc}(x) = \sin(x)/x$ captures the intuition that atoms far apart interfere less strongly. Because every pair must be evaluated, this scales as $O(N^2)$ in the number of atoms. The Stuhmann decomposition (Eq. 2) expands the Debye equation into a sum of spherical harmonics, decomposing the molecular shape into a sum of L^2 angular modes, truncated at order L . This reduces the time complexity down to $O(N \cdot L^2)$, and assuming for most cases $L \ll N$, reduces to $O(N)$ [7]. This project uses the in-vacuo form (no solvent, hydration shell, or intermolecular forces); a live demo is available online.

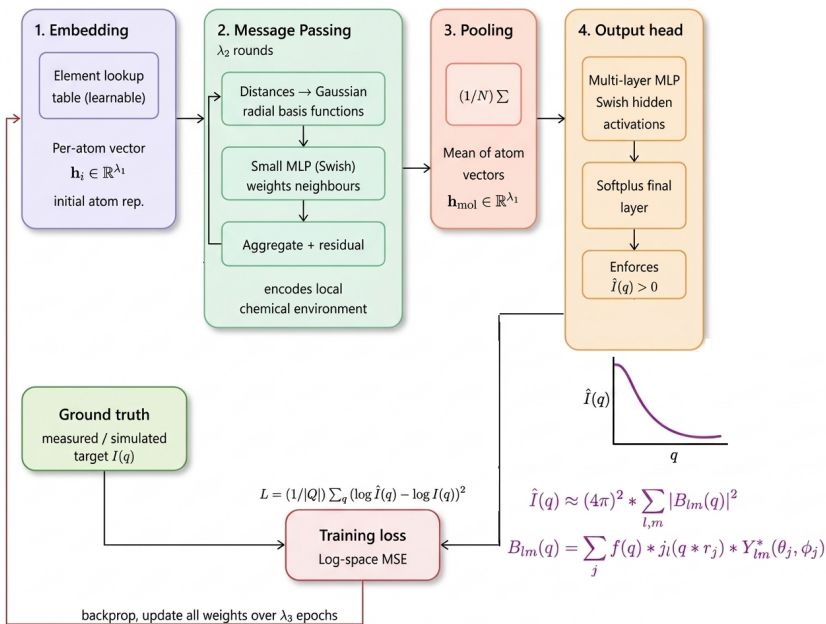
$$B_{lm}(q) = \sum_i f_i(q) j_l(qr_i) Y_{lm}^{*}(\theta_i, \phi_i), \quad I(q) = (4\pi)^2 \sum_{l,m} |B_{lm}(q)|^2 \quad (2)$$

2 Data Processing

A dataset of **1,139,747 molecules** was assembled from 13 sources with diverse composition. In total, 474 unique atom/ion types are represented (charge-resolved, e.g. 0 and 02- are distinct). Full source attributions are provided in `sources.tsv`, embedded in the database. Raw structures were obtained in various formats (.cif, .pdb, .xyz) and converted to a uniform Cartesian XYZ representation. Dummy and non-physical atom sites were removed, deuterium was mapped to hydrogen, and structures declaring zero atoms were discarded. Non-element placeholders (X, M, Cn, etc.) were excluded from the ionic makeup. $I(q)$ ground truths were then computed via the Stuhmann decomposition (source code) at $L=50$ and stored in a compressed HDF5 database.

3 Proposed Architecture

The proposed network takes a molecule’s atomic positions and element types as input and predicts $\hat{I}(q) \in \mathbb{R}^{|Q|}$. Let hyperparameters λ_1 be the hidden dimension, λ_2 the number of message-passing rounds, and λ_3 be the number of training epochs. **1) Embedding:** Atomic coordinate files are converted into a list of atoms, which are looked up in a trainable table, producing a vector $\mathbf{h}_i \in \mathbb{R}^{\lambda_1}$. This gives the network a learnable representation of each element. **2) Message passing (λ_2 rounds):** Each atom updates its vector by aggregating information from nearby neighbours. Interatomic distances are encoded as Gaussian radial basis functions before being passed to a small MLP (with



Swish activations) that weights each neighbour’s contribution. A residual connection is added at each round. After λ_2 rounds, each atom’s vector encodes its local chemical environment. **3) Pooling:** Atom vectors are averaged into a single fixed-size molecule vector $\mathbf{h}_{\text{mol}} \in \mathbb{R}^{\lambda_1}$, independent of molecule size. **4) Output head:** A multi-layer MLP with Swish hidden activations maps $\mathbf{h}_{\text{mol}}$ to $\hat{I}(q) \in \mathbb{R}^{|Q|}$. A Softplus activation on the final layer enforces strict positivity $\hat{I}(q) > 0$, consistent with the physical requirement that scattering intensity is always positive. **5) Backprop:** The model is trained a total of λ_3 epochs against ground truths. The network is trained to minimise a log-scale MSE loss, $\mathcal{L} = \frac{1}{|Q|} \sum_q (\log \hat{I}(q) - \log I(q))^2$ which accounts for the orders-of-magnitude variation in intensity across the dataset.

4 Baseline Model

The baseline is a small MLP that takes as input only the atom-type histogram of a molecule (i.e. a count vector $\mathbf{c} \in \mathbb{Z}^{|\mathcal{E}|}$ recording how many atoms of each element type are present) and maps it directly to $\hat{I}(q) \in \mathbb{R}^{|Q|}$ via two hidden layers with Swish activations and a Softplus output. The model sees no atomic coordinates, no interatomic distances, and no graph structure whatsoever.

5 Ethical Considerations

All training data is open-source and fully attributed; no private or sensitive information is involved. The primary risk is misuse of model predictions: since the GNN is an approximation, predicted $I(q)$ curves could mislead structural biology or materials research if treated as exact without cross-validation against the ground-truth computation.

References

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